FEDERAL SPECIFICATION

TESTER, DENSITY AND MOISTURE (SOIL AND ASPHALT), NUCLEAR METHOD

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers a one piece (gamma and neutron) source moisture and density tester.

1.2 Classification.

Type I - Direct transmission, backscatter, thermal neutron moisture.
Type II - Backscatter/air gap, thermal neutron moisture.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Military Standard:

MIL-STD-1458 - Radioactive Materials: Marking and Labeling of Items, Packages and Shipping Containers for Identification in Use, Storage and Transportation.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

CFR 49 170-179 - Transportation, Parts 170-179.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 2922 - Density of Soil and Soil Aggregate in Place by Nuclear Methods (Shallow Depth).
D 2950 - Density of Bituminous Concrete in Place by Nuclear Method.
D 3017 - Moisture Content of Soil and Soil Aggregate in Place by Nuclear Methods (Shallow Depth).

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3. REQUIREMENTS

3.1 Description. The tester shall be a single unit radiographic device containing a radiation gamma and neutron source. The tester shall be self-contained and shall consist of two basic electronic systems: (a) A radioactive source, radiation detectors, and related electronic components; (b) A scaler system containing power supplies, counting circuits, timing circuits, data display, and related electronic components. The measurement of moisture shall be by the principle of thermalization of fast neutrons. The measurement of density shall be by the principle of direct transmission and backscatter, type I, or backscatter/air gap, type II.

The tester shall be the latest model of the standard product of the supplier and shall have been used in the field for at least 1 year by professional organizations such as State Highway Departments. The introduction of normal product improvement changes in this 1-year period is acceptable. Standard or optional equipment, as listed in the supplier's published brochures and catalogs or as offered by the supplier, shall be furnished to meet the requirements specified herein. When specified requirements are not stated herein, all items listed as standard equipment, or normally furnished as standard equipment, shall be furnished. The tester shall conform to all Federal laws and regulations governing safety for this type of equipment.

3.2 Electrical operation. The tester shall be capable of operation from the power sources specified in (a) and (b) and have the electrical design characteristics specified in (c), (d), (e) and (f) as follows:

(a) An internal, rechargeable (nickel cadmium) battery pack capable of 300 consecutive measurements in the 60-second mode, without recharge.

(b) Any external 110 volt, 60 cycle, AC source (type I only).

(c) A battery status indicator.

(d) A low battery voltage cutoff to protect the battery from deep discharge.

(e) All user serviceable circuitry, including the battery pack, shall be plug-in type, modular construction and solid state design. The tester shall accept ordinary alkaline cells for temporary emergency use.

(f) All circuitry shall be operated as a complete unit for a "burn-in" period of not less than 30 hours.

3.3 Temperature conditions. The tester shall operate as specified herein at any ambient temperature from 0°F to + 125°F. The tester shall operate as specified herein on hot materials (asphalt) at temperatures up to + 350°F without the use of insulation materials between the instrument and the surface.

3.4 Safety. The radiation sources used shall be a sealed source type construction of 137 Cesium and 241, Americium/Beryllium, type I, or 226 Radium/Beryllium, type II. The radiation level at 30.48 centimeters (12 in) from any surface with the source in the shielded position shall not exceed 10 millirems per hour (mrem/hr). The source shall meet the leak testing requirement of less than 0.005 microcuries (uci). The shipping container of the tester shall meet the Radioactive Yellow II labeling requirements for ground transport without a container.

3.5 Licensing for radioactive material. The supplier shall certify that the tester has met the product safety requirements of the F.D.A. Bureau of Radiological Health, for Radium 226, or the U.S. Nuclear Regulatory Commission, for 137 Cesium and 241 Americium/Beryllium as appropriate and that the necessary licensing data has been provided. The contracting officer will furnish the supplier a copy of the license prior to shipment date required.

3.6 Performance. The tester shall meet the performance requirements of ASTM Standards D-2922, D-2950, and D-3017. The tester shall also meet the following requirements as appropriate:

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Direct Transmission - Type I

Range of calibration	80 to 160 pounds per cubic foot (pcf)
Precision (6 in depth) (app. 120 pcf).	± 0.25 pcf.

Backscatter - Type I

Range of calibration.	80 to 160 pcf.
Precision (app. 120 pcf.)	± 0.50 pcf.

Backscatter/Air gap - Type II

Range of calibration.	80 to 170 pcf.
Precision (app. 120 pcf).	± 1.00 pcf.

Moisture - Types I and II

Range of calibration.	0 to 40 pcf.
Precision (app. 15 pcf).	± 0.35 pcf (standard deviation for 20, 1 minute measurements).

3.7 Asphalt content. Under conditions of known asphalt water content, the tester shall be capable of being calibrated to measure the asphalt content of an asphaltic concrete mix.

3.8 Scaler system.

3.8.1 Timer. The timer shall be a quartz crystal-controlled type with a precision better than 50 parts per million and at least one interval that meets the requirements of section 3.6.

3.8.2 Voltmeter. The scaler shall have a device to indicate whether the battery voltage is adequate for testing.

3.8.3 Circuitry. The circuitry shall be of module type. The circuitry shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish or equal.

3.8.4 Readout. The readout shall be a liquid crystal display (LCD), type I, or light emitting diode (LED), type II, which is easily readable without shading in high or low ambient light conditions. The readout shall be capable of giving a count at the end of a preset time interval. The substitution of a ratemeter will not be accepted. The count for type I shall be a direct readout in wet density, dry density and moisture in PCF or Standard International Units and in percent moisture, or for type II convertible to PCF or Standard International Units by using the proper calibration tables. The count of type II may be specified as direct readout in wet density, dry density and moisture in PCF or Standard International Units and in percent moisture (see 6.2).

3.9 Source system.

3.9.1 Source. The source shall be encapsulated in a manner so that it is in compliance with applicable welding in federal and state guidelines. The sealed source shall be contained by fusion welding in a source rod having a minimum hardness of 60 as rated on the Rockwell B scale. Type I only sealed source shall be permanently positioned within the source rod on source holder. Either lead or carbide-tungsten shall be used to comply with the radiation shielding requirements specified in 3.4.

3.9.2 Detectors. The density detector shall consist of one or more Geiger-Muller tubes. The moisture detector shall consist of one or more borontrifluoride or Helium-3 tubes.

3.10 Air gap stand (type II only). The air gap stand shall be an integral part of the tester and selection of the airgap position shall be made via the source position control and, further, an electronic sensing of position selected shall be made to insure proper data collection.

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3.11 Source mechanism. The source mechanism shall provide direct position indication of source position and a keyed lock for the shielded position shall be provided.

3.11.1 Transmission probe (rod) - (type I). The rod shall be equipped with a keyed locking device to prevent accidental unshielding of the source. The construction of the rod shall be such that soiling of the rod, resulting from usage, shall not cause sticking. The rod shall have permanent markings in 2-inch increments to denote, to not less than 8 inches, the depth below ground of the transmission readings. An inspection plate below the rod and the shield shall be provided. This plate shall have means to clean the rod while being retracted.

3.11.2 Depth control (type II). A depth control shall be provided to permit the operator to change the depth of test from full depth, 6 inches minimum, to shallow depth, 1 inch, for thin concrete or asphaltic concrete overlays on bridge decks and other surfaces.

3.12 Weight. The tester shall weigh not more than 50 pounds. The tester and case (with accessories) shall weigh not more than 120 pounds.

3.13 Battery charger. The tester shall have an automatic charger. The charger shall be connected with plug-in type quick disconnects. The tester shall incorporate a battery overcharge prevention feature. The charger shall operate from power sources of 105-130 volts, and may operate from additional power sources of 12 to 18 volts DC and 210 to 250 volts, 47 to 63 Hz, AC (see 6.2).

3.14 Construction. The construction, materials, and finish of items supplied shall be normal commercial practice suitable to the use of the equipment.

3.15 Identification plates and marking. Each tester shall be identified in accordance with the supplier's standard method. Each tester shall be serialized, with each instrument and its associated density and moisture standards, air gap stand, and all data marked identically.

3.15.1 Radiation marking. The marking, on the tester, shall include the radiation caution symbol, quantity of isotope, date (see 3.5), and appropriate control inspections of the using service (see 6.2). A warning against tampering with the shielding and the source within the tester shall also be included.

3.16 Maintainability. The tester shall be modular and designed for maximum ease of maintenance consistent with reliability, electrical safety, and radiological safety. All user serviceable subassembles shall be removable without exposing the source or disturbing the shielding assembly. The scaler subsystem and calibrated source/detector subsystem shall be readily removable for interchange with identical type subsystems.

3.17 Technical publications. Unless otherwise specified (see 6.2), two sets of the supplier's standard operating, maintenance, safety, and parts manuals shall be furnished with each tester.

3.18 Transport case. A transport case constructed of metal, wood, or plastic shall be furnished. Shock-absorbing material or mounting for all devices shall be provided within the case. The case shall contain the tester and all tools and accessories, reference standards, technical publications, and all data. The case shall have at least two handles and locking devices. The hardward shall be corrosion-resistant. The transport case shall meet all regulations in CFR 49 170-179. The case shall include papers certifying that the package design meets CFR requirements.

3.19 Reference standard. A reference standard shall be furnished. It shall be a combination type provided for the purpose of checking equipment operation, background count, and to establish conditions for a reproducible density and moisture count rate.

3.20 Lubricants. The tester shall operate as specified herein when lubricated with standard commercial lubricants.

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3.21 Paint. Commercial painting shall be provided. It shall include the supplier's commercial cleaning, preparation, and priming procedures. Color for the radiological signs, labels, and symbols shall comply with Title 10 and Title 49 of the Code of Federal Regulations.

3.22 Tools and accessories. The following items shall be furnished with each tester:

	Item	Quantity
(a)	Cables, charger - as required.	Varies with the manufacturer.
(b)	Site preparation device (type I) - A site preparation device, with handles, shall be provided for purpose of preparing a level test and for vertical alignment of the drill rod for transmission hole preparation.	1
(c)	Drill pin (type I) - A hardened steel for preparing a hole normal to the surface for transmission measurements. It shall not exceed the diameter of the source rod by more than 1/8".	1
	Sliding Hammer (type I - (Optional) - A hardened steel drill pin with integral sliding driving and retraction hammer for preparation of a hole normal to the prepared surface for transmission measurements. It shall not exceed the diameter of the source rod by more than 1/8". To be purchased as optional item. NOTE: Not all type I manufacturers supply this item (see 6.2).	1
(d)	Site preparation plate (type II) - A site preparation plate with handles shall be provided for the purpose of preparing a flat level test site.	1

3.23 Calibration data. Calibration charts, where required, for all applicable operating modes for both moisture (in 0.25 pcf increments 0 to 45 pcf) and density (in 0.5 pcf increments 80 to 160 pcf) shall be provided. These calibration charts shall be applicable to all normal construction type soil, aggregate, and pavement mixtures in ranges specified without the need for field correction for material composition.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Quality conformance inspection (see 4.4).

4.3 Examination. Prior to demonstrating, the tester shall be examined for the defects in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.1 Demonstrations. The tester shall be tested in accordance with 4.5.2.1 through 4.5.2.4. Failure of any test shall be cause for rejection.

4.3.2 Examination of preparation for delivery. An examination shall be performed to determine compliance with the packing, palletization (when supplied) and marking requirements of section 5. The sample unit shall be one container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

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4.4 Quality conformance inspection.

4.4.1 Examination. Each tester shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Demonstrations. Each tester shall be demonatrated in accordance with 4.5.2.3. Failure of the demonstration shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The tester shall be examined for the following defects:

101. Electrical operation not as specified.
102. Radiation source not as specified.
103. License or control of radiation source not as specified.
104. Scaler system not as specified.
105. Probe system not as specified.
106. Air gap stand not as specified (optional).
107. Source mechanism of transmission probe (rod) not as specified.
108. Weight not as specified.
109. Battery charger not as specified.
110. Construction not as specified.
111. Identification plates and marking not as specified.
112. Maintainability not as specified.
113. Technical publications not as specified.
114. Transport case not as specified.
115. Reference standard not as specified.
116. Lubrication not as specified.
117. Paint not as specified.
118. Tools and accessories not as specified.
119. Calibration data not as specified.

4.5.2 Demonstration.

4.5.2.1 Operating. The tester shall be operated at 0°F and + 125°F, and on a surface having a temperature of 350°F. Nonconformance to 3.3 shall constitute failure of this demonstration.

4.5.2.2 Safety. The radiation limits of the tester shall be measured in accordance with radiological safety regulations. Nonconformance to 3.4 shall constitute failure of this demonstration.

4.5.2.3 Precision. The tester shall be operated in accordance with the manufacturer's instructions to obtain 10 consecutive measurements in each of its operating modes. Nonconformance with paragraph 3.6 shall constitute failure to meet this requirement. For this evaluation the following definitions apply:

Precision shall be the calculated standard deviation of each set of 10 consecutive measurement values.

Accepted value (of reference samples) shall be determined as follows:

Wet or bulk density shall be the ratio of measured net weight to measured volume of the entire sample. (See ASTM 2922, Density of Soil and Soil Aggregate in Place by Nuclear Methods (Shallow Depth.)

Moisture content shall be the difference between the wet and dry weights of an oven dried sample of not less than 0.1 cubic foot taken from the center of the sample as a cylindrical section extending the full depth of the sample. (See ASTM 3217, Moisture Content of Soil Aggregate in Place by Nuclear Methods (Shallow Depth).

4.5.2.4 Under conditions of known asphalt water content, it shall be demonstrated that the tester is capable of being calibrated to measure the asphalt content of an asphaltic concrete mix. Nonconformance to 3.7 shall constitute failure of this demonstration.

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5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. The assembled tester, tools, accessories and technical publications of one type only, or component parts as specified (see 6.2) shall be preserved and packed in a box conforming to PPP-B-636, weather resistant. Interior packing forms, scored and folded, molded or die-cut type shall be used to suspend, cushion and provide protection against damage during shipment, handling and storage. The box shall be closed, waterproofed and banded in accordance with the appendix to PPP-B-636.

5.1.2 Commercial. The assembled tester, tools, accessories, and technical publications of one type only, or component part, as specified (see 6.2) shall be preserved and packed in a fiberboard box in accordance with normal commercial practice. The complete pack shall be designed to protect the item against damage during shipment, handling and storage, to insure delivery at destination, to provide for redistribution by the initial receiving activity and be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.2 Palletization. When shipments to Government depots are full car or truckload, the shipping containers shall be palletized on wooden pallets, 2 or 4 way design, to facilitate handling in accordance with normal commercial. The palletized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.3 Marking. Marking shall be as specified in the contract or order (see 6.2).

6. NOTES

6.1 Intended use. The tester is used for measurement of density and moisture content of soils, base materials, and density of asphaltic concrete.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of tester required (see 1.2).
(c) Licenses and authorizations issued to the using service for control of the radiation source (see 3.5).
(d) Whether direct readout for wet density and moisture is required for type II tester (see 3.8.4).
(e) Battery charger operation from 12 to 18 volts DC and/or 210 to 250 volts AC required (see 3.13).
(f) Radiation control instructions required by the using service (see 3.15.1).
(g) Technical publications required if other than as specified (see 3.17).
(h) Whether sliding hammer required for type I tester (see 3.22).
(i) Packing and marking required (see 5.1 and 5.3).

Review Interest:

Army - ME

Preparing Activity:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 60 cents each.

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